

Certified analytical bounds for hypersonic reachable sets

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Abstract

This is a proposal in nonlinear analysis and dynamical systems. The object is a singularly perturbed family of differential inclusions in which compactness fails along the time axis: as the parameter vanishes, the motion that carries the important state transitions occupies a vanishing time, so it disappears from the ordinary limits of the family while continuing to decide where the trajectories go. The remedy we propose is a *profile decomposition* — a low-dimensional *skeleton*, finitely many concentrating *bubbles*, and a *residual* bounded in a stated metric — and the question is whether that decomposition is a theorem about solution sets rather than a description of them.

We state the question as a hierarchy. *Theorem A*: every sequence of trajectories splits, along a subsequence, into a skeleton, finitely many rescaled profiles, and a remainder. *Theorem B*: the profiles induce a set-valued jump map. *Theorem C*: the solution set of the family is contained in that of the skeleton-and-jump system, thickened by an explicit ball. This document proves Theorem C, with every constant written down, for a model whose potential is asymmetric in the one way that produces concentration, and whose profiles are counted *a priori* by a Lyapunov functional rather than extracted by an iterated dichotomy. That is where we expect the contribution to the analysis to lie: the classical concentration-compactness argument yields the structure by contradiction and produces no constant, whereas a dissipation functional yields the same structure quantitatively.

Hypersonic atmospheric entry is the example that supplies the problem and the one against which the result is to be measured; it is not the subject. [Section 7](#) states the correspondence and what the full system adds. Only the theorem of [Section 4](#) is claimed as proved.

1 The question, reduced

The object we want enclosed is a reachable set: every state a control system can be driven to, under every admissible control and every admissible model of its forcing. For an atmospheric entry vehicle this is what a safety analysis needs, and two things stand in the way — the dimension of the state, and the provenance of the guarantee, since a bound produced by a numerical solver inherits that solver’s accuracy. Our program addresses both, and [Section 7](#) describes it. This document does something narrower and prior to it: it takes the one analytical difficulty on which the program turns, strips it of the application, and proves a theorem about it. It addresses the dimension and sets the provenance aside ([Section 1.1](#)).

The difficulty is a singular limit of a kind that defeats the ordinary compactness arguments. The example supplies the parameter: for atmospheric entry it is the ratio $\varepsilon = H_s/R_e$ of the atmospheric scale height to the planetary radius, whose square root is the ratio of the period of the oscillation about equilibrium to the slow time of the flight. As $\varepsilon \rightarrow 0$ the motion separates into arcs on which the forcing is not felt and *passes* in which it is felt strongly for a vanishing time. A pass becomes instantaneous and keeps its effect, so in the limit a trajectory is a sequence of arcs joined by jumps — and yet the natural limiting objects do not see the jumps, because they weigh an interval of time by its length and a pass has none. A solution set is a support and not an integral, so what has no mass still counts.

This is the structural phenomenon of concentration, and the language for it is a profile decomposition: a low-dimensional *skeleton*, which is what survives the limit; one *bubble* per pass, the pass rescaled to its own width, whose net effect on the surviving coordinates is a jump; and a *residual*, which is what the skeleton and the finitely many bubbles do not capture. In the theory of nonlinear dispersive equations such a decomposition is obtained by iterating the dichotomy of the concentration-compactness principle [1, 2], and what it delivers is structure: it says which pieces can occur and that nothing else can, by contradiction, and it produces no constant. The authors of the wave-equation theory are explicit that the uniform bound following from their decomposition carries a constant they cannot describe [2, Corollary 2].

Our proposal is that here the structure can be had *quantitatively*, because the family carries a Lyapunov functional. The dissipation along a trajectory is monotone and its total variation is bounded uniformly in the parameter, so the number of profiles above any threshold is bounded in advance rather than extracted by a contradiction argument, and the same functional bounds the time the motion may spend concentrating and the accumulated growth of the comparison. We therefore state the question as the hierarchy of the abstract — the decomposition, the jump map it induces, and the inclusion for solution sets — and this document proves the third for a model that keeps the concentration and discards everything else.

Remark 1.1 (What the small parameter is, and what it is not). It is tempting to write the reduced problem as a slow state driven by a fast one, $\dot{z} = g(z) + h(y)$ with $\varepsilon \dot{y} = q(y, z)$ and h bounded. That form does not contain the difficulty: if the excursion in y lasts a time of order ε and the forcing is of order one, its contribution to z is of order ε , so in the limit it leaves no trace and ordinary averaging describes everything. What produces a jump is not a large forcing imposed by hand but an *asymmetric potential*. In the model of [Section 2](#) the air supplies a wall that steepens without bound below the equilibrium while gravity is uniform above it, so a large oscillation rebounds in a short time and coasts for a long one. The ratio of the two is set by the amplitude alone ([Lemma 3.4](#)), and ε — the square of the ratio of the oscillation period to the slow time — only decides how fast the orbit is traversed. Concentration is the asymmetry of the well; it is not an assumption about the size of a force.

1.1 What this document does not prove

[Theorem 4.2](#) is Theorem C of the hierarchy, for one model. Three further statements belong to the programme and are not proved here; we name each with the role it plays, so that what is missing is visible rather than implied.

Theorem A: the decomposition itself. That every sequence of trajectories with $\varepsilon_n \rightarrow 0$ splits, along a subsequence, into a skeleton, finitely many profiles obtained by rescaling the passes to their own widths, and a remainder. [Theorem 4.2](#) compares a trajectory with a trajectory of the limit system and does not assert this; what the Lyapunov functional of [Assumption 2.3](#) supplies is the bound on the number of profiles that the classical argument obtains by contradiction, so the decomposition is expected to follow quantitatively. Proving it is [objective 1](#).

The limit of the family in the weak topology. The dynamics become a linear constraint when the family is embedded in a space of occupation measures, and the solution set is then the support of a terminal measure. [Section 6](#) gives the definitions and the topology and states what the companion to [Theorem 4.2](#) would assert at that level; proving it is [objective 4](#).

The passage from the weak limit back to the support. A solution set is a support and not an integral. Granting the limit of the measures and the profiles it carries, that the limiting solution set is contained in the one those profiles generate is a further implication, and the hardest link in the chain; the proof below obtains the inclusion from trajectories directly and so does not cross it. Which route the general case should take is open, and it is the question we would most value an advisor's judgment on.

One further obligation is of a different kind and is deferred rather than open. A bound in closed form still has to be evaluated, and evaluation in floating point is where a guarantee is ordinarily lost; the setting in which it is not is validated interval arithmetic, in which every operation is rounded outward so that the computed interval encloses the exact result [3, 4, Ch. 3]. That is a question about evaluating a bound rather than about proving one, and it returns as [objective 5](#).

1.2 Terminology

Two fields meet here and several words mean different things in each. We fix them once.

term	what it means here
<i>orbit</i>	One closed curve of Eqs. (2.1) to (2.2) in the (z, v) plane: a single complete oscillation, in the sense of Lemma 2.7. <i>Not</i> a trajectory about the planet — the motions considered here are sub-orbital and decelerating throughout. A skip is one orbit; a multi-pass entry is a succession of them, with the amplitude decaying in between.
<i>pass, dip</i>	The part of an orbit below the section $z = 0$.
<i>bubble</i>	A pass rescaled to its own width: a concentrating profile, whose net effect on the slow coordinates is a jump.
<i>skeleton</i>	What survives the limit: the arcs between passes, and the glide.
<i>residual</i>	The ball by which the inclusion is thickened.
<i>amplitude W</i>	The energy $\frac{1}{2}v^2 + \Phi_{c,\theta}(z)$ of the oscillation, and a Lyapunov functional for it. Not an amplitude in the sense of a maximum displacement.
<i>averaging</i>	Always over one orbit, with a single phase (Lemma 4.4). Averaging over <i>several</i> phases, whose frequencies may cross, is a different problem, is not attempted anywhere below, and is the subject of objective 3.
<i>the oscillator</i>	Always the motion of Eqs. (2.1) to (2.2) — the altitude about the glide. The <i>attitude</i> oscillator of Section 7, three orders of magnitude faster, is named in full wherever it appears.

2 The model and its hypotheses

Fix an integer $d \geq 1$ and a horizon $T > 0$. For $\varepsilon \in (0, 1]$ the state is $x = (z, v, s)$, with $z \in \mathbb{R}$ an altitude measured in layer thicknesses, $v \in \mathbb{R}$ its rescaled rate, and $s \in \mathbb{R}^d$ a vector of slow coordinates. The system is

$$\sqrt{\varepsilon} \dot{z} = v, \quad (2.1)$$

$$\sqrt{\varepsilon} \dot{v} = c m_\theta(z) - 1 - \sqrt{\varepsilon} \kappa(s, c, \theta) v, \quad (2.2)$$

$$\dot{s} = F(s, c, \theta) + m_\theta(z) \Gamma(s, c, \theta). \quad (2.3)$$

Here $c(\cdot)$ is the control, $\theta(\cdot)$ is a selection — a choice of admissible field from a bounded family — and m_θ is a density shape. Writing M_θ for the antiderivative of m_θ with $M_\theta(0) = 0$, the *potential* is

$$\Phi_{c,\theta}(z) = z - c M_\theta(z), \quad \text{so} \quad \Phi'_{c,\theta}(z) = 1 - c m_\theta(z), \quad (2.4)$$

and Eqs. (2.1) to (2.2) are a unit-mass particle in that potential with a weak damping. The *amplitude* is its energy,

$$W = \frac{1}{2}v^2 + \Phi_{c,\theta}(z). \quad (2.5)$$

The whole of Section 3 is an account of one object: the shape of this potential.

Assumption 2.1 (H1: the wall). For each $\theta \in \Theta$ the shape $m_\theta \in C^1(\mathbb{R})$ satisfies $m_\theta > 0$ and $m'_\theta < 0$; $m_\theta(z) \rightarrow 0$ as $z \rightarrow +\infty$ with $\int_0^\infty m_\theta < \infty$; $m_\theta(z) \rightarrow \infty$ as $z \rightarrow -\infty$; and $1/c$ lies in the range of m_θ for every $c \in [c_{\min}, c_{\max}]$. The bounds $0 < m_- \leq m_\theta \leq m_+$ and $|m'_\theta| \leq m_1$ hold uniformly over Θ on the corridor of Assumption 2.5.

The two one-sided conditions are the whole physical content: the air thins upward without bound on the integral, and thickens downward without limit. Nothing in what follows uses the shape of m_θ beyond Assumption 2.1, which is why θ may select it: an enclosure of an uncertain density is exactly a bounded family of admissible shapes.

Assumption 2.2 (H2: data). $[c_{\min}, c_{\max}] \subset (0, \infty)$, Θ and $S \subset \mathbb{R}^d$ are compact. The maps $F, \Gamma: S \times [c_{\min}, c_{\max}] \times \Theta \rightarrow \mathbb{R}^d$ and κ are continuous, bounded by $F_{\max}$, $\Gamma_{\max}$ and $\kappa_{\max}$, and Lipschitz in s uniformly in (c, θ) with constants κ_F, κ_Γ . Moreover $\kappa \geq \kappa_{\min} > 0$.

Assumption 2.3 (H3: two Lyapunov functionals). There are $\Theta_1, \Theta_2 \in C^1(\mathbb{R}^d)$ and constants $c_1, c_2 > 0$ with

$$\nabla \Theta_1 \cdot F \leq 0, \quad \nabla \Theta_1 \cdot \Gamma \leq -c_1, \quad \nabla \Theta_2 \cdot F \leq 0, \quad \nabla \Theta_2 \cdot \Gamma \leq -c_2 \sqrt{m_\theta} / m_\theta,$$

for all (s, c, θ) . Write $M = (\max_S \Theta_1 - \min_S \Theta_1) / c_1$ and $Q = (\max_S \Theta_2 - \min_S \Theta_2) / c_2$.

Two Lyapunov functionals for the slow motion, weighted differently in the density. Each is non-increasing along every trajectory, for every control and every selection, and the bound on its total variation over the compact corridor is what converts monotonicity into a count. The first is the energy identity of the application, and it bounds $\int m_\theta d\tau$; the second is weighted by $\sqrt{m_\theta}$ and bounds $\int \sqrt{m_\theta} d\tau$. They cut in opposite directions, which is the point: for a fixed value of the first integral, concentrating the density *lowers* the second, so the first bounds how many deep excursions there can be and the second bounds how long the motion may linger at moderate density. In the application the two are the energy to be dissipated and the integrated heat load.

Assumption 2.4 (H4: control). $c(\cdot)$ is absolutely continuous with values in $[c_{\min}, c_{\max}]$ and $|\dot{c}| \leq \dot{c}_{\max}$. (The sharper form, in which c is piecewise constant on intervals of length at least Δ , is [Remark 3.2](#).)

Assumption 2.5 (H5: corridor). Every trajectory remains on $[0, T]$ in a compact corridor $\mathcal{X} = \{|z| \leq z_{\max}, |v| \leq v_{\max}, s \in S\}$, and $x(0) \in \mathcal{X}_0 \subseteq \mathcal{X}$.

Remark 2.6 (What is uniform, and what the hypotheses are not). Every constant named above is a bound over $S \times [c_{\min}, c_{\max}] \times \Theta$. No constant below depends on the particular control, the particular selection, or the particular initial point: that is the sense in which the result is uniform over the admissible family, and it is why an enclosure of an uncertain field is the natural setting rather than a complication. Note also what is absent. There is no transversality condition, no non-grazing condition, and no assumption that a pass is a single dip — the last is [Lemma 2.7](#), a consequence of $m'_\theta < 0$. Every hypothesis is a condition on the data, checkable before any trajectory is computed.

Lemma 2.7 (One well, and one dip per period). *Under [Assumption 2.1](#) the potential $\Phi_{c,\theta}$ is strictly convex, with*

$$\Phi''_{c,\theta}(z) = -c m'_\theta(z) > 0, \tag{2.6}$$

and attains its minimum at the unique $z_(c, \theta)$ where $c m_\theta(z_*) = 1$. Consequently every sublevel set $\{\Phi_{c,\theta} \leq W\}$ is a bounded interval $[z_-(W), z_+(W)]$, and at frozen (c, θ) every bounded motion of [Eqs. \(2.1\) to \(2.2\)](#) with $\kappa = 0$ is a single closed orbit — one closed curve in the (z, v) plane, traversed periodically — on which z attains exactly one minimum and one maximum.*

Proof. Differentiating [Eq. \(2.4\)](#) twice gives [Eq. \(2.6\)](#), which is positive because $c > 0$ and $m'_\theta < 0$. A strictly convex C^2 function has at most one stationary point; $\Phi'_{c,\theta} = 1 - c m_\theta$ changes sign exactly once because m_θ is strictly decreasing from $+\infty$ to 0 and $1/c$ is in its range, so z_* exists and is unique. Strict convexity makes each sublevel set an interval, bounded because $\Phi_{c,\theta} \rightarrow +\infty$ at both ends. At frozen (c, θ) and $\kappa = 0$ the quantity [Eq. \(2.5\)](#) is constant, so the orbit is the level curve $\{\frac{1}{2}v^2 = W - \Phi_{c,\theta}(z)\}$, which is a single closed curve meeting $v = 0$ exactly at the two endpoints of the interval. Those are the one minimum and the one maximum of z . $\square$

[Lemma 2.7](#) is worth noting for what it is not: a hypothesis. That a pass is a single dip rather than a train of them is a property of the motion, and here it costs one sign on the density rather than a condition imposed on solutions.

3 Three regimes of one oscillator

Glide, phugoid and skip are not three systems but three energies in the single well of [Lemma 2.7](#). This section says which is which, and the statement that organises everything is that ε sets no shape — only a clock.

Lemma 3.1 (The amplitude law). *Along every trajectory of [Eqs. \(2.1\) to \(2.3\)](#),*

$$\dot{W} = -\kappa(s, c, \theta) v^2 - M_\theta(z) \dot{c}, \tag{3.1}$$

whatever the shape of m_θ . Consequently, with $M_* = \max_{\mathcal{X}, \Theta} |M_\theta|$,

$$W(\tau) \leq W(0) + M_* \dot{c}_{\max} \tau \quad \text{for } \tau \in [0, T]. \quad (3.2)$$

Proof. Differentiate Eq. (2.5) along Eqs. (2.1) to (2.2):

$$\dot{W} = v\dot{v} + \Phi'_{c,\theta}(z)\dot{z} + \frac{\partial \Phi_{c,\theta}}{\partial c} \dot{c} = \frac{v(c m_\theta - 1)}{\sqrt{\varepsilon}} - \kappa v^2 + \frac{(1 - c m_\theta)v}{\sqrt{\varepsilon}} - M_\theta(z)\dot{c},$$

where $\partial \Phi_{c,\theta}/\partial c = -M_\theta(z)$ by Eq. (2.4). The two terms carrying $\varepsilon^{-1/2}$ cancel identically, which is Eq. (3.1); the cancellation uses only that the same $c m_\theta$ appears in Eq. (2.2) and in $\Phi'_{c,\theta}$. Since $\kappa > 0$, Eq. (3.1) gives $\dot{W} \leq M_* \dot{c}_{\max}$, and Eq. (3.2) follows by integration. $\square$

So W is a Lyapunov functional for the fast motion, decaying except where the control pumps it, and Assumption 2.4 is what makes Eq. (3.2) an *a priori* bound rather than a hypothesis about solutions. Together with the two functionals of Assumption 2.3 — one for the slow motion weighted by the density, one by its square root — these are the three monotone quantities on which every count below rests.

Remark 3.2 (The sharper form, from a finite guidance interval). If c is piecewise constant on intervals of length at least Δ — which is what a finite guidance interval supplies — then $\dot{c} = 0$ between switches and Eq. (3.1) reads $\dot{W} = -\kappa v^2$ exactly: W is non-increasing on each interval, and changes only by the jumps $|\Delta W| \leq M_* |\Delta c|$ at the at most T/Δ switching times. Equation (3.2) then holds with $M_* \dot{c}_{\max} \tau$ replaced by $M_* |\Delta c|_{\max} \lceil \tau/\Delta \rceil$, and LaSalle's argument applies interval by interval.

3.1 The orbit averages, and the time the oscillation may occupy

Throughout, a *orbit* is one closed curve of Eqs. (2.1) to (2.2) in the (z, v) plane — a single complete oscillation, in the sense of Lemma 2.7 — and never a trajectory about the planet. A skip is one orbit; a multi-pass entry is a succession of them, with the amplitude decaying between. Two exact identities do most of the quantitative work, and both come from periodicity rather than from any property of m_θ .

Lemma 3.3 (Orbit averages). *Let $\langle \cdot \rangle$ denote the time average over one closed orbit of Eqs. (2.1) to (2.2) at frozen (c, θ) and $\kappa = 0$. Then*

$$\langle m_\theta \rangle = \frac{1}{c}, \quad (3.3)$$

for every amplitude W and every shape satisfying Assumption 2.1. Consequently the impulse of a orbit equals its period divided by c , and along any trajectory of Eqs. (2.1) to (2.3) the total time spent in closed oscillation satisfies

$$\sum_{\text{orbits}} (\text{period}) \leq c_{\max} M. \quad (3.4)$$

Proof. Integrate Eq. (2.2) over one period at $\kappa = 0$. The left side is $\sqrt{\varepsilon}[v] = 0$ because v is periodic, so $\int (c m_\theta - 1) d\tau = 0$, that is $c \int m_\theta d\tau = \int d\tau$, which is Eq. (3.3) on dividing by the period. The impulse of the orbit is $\int m_\theta d\tau = \langle m_\theta \rangle \times (\text{period}) = (\text{period})/c$. Summing over orbits and using $\int_0^T m_\theta d\tau \leq M$ from Assumption 2.3 gives Eq. (3.4). $\square$

Equation (3.3) deserves a word, because it is the reason the skeleton is the glide. Whatever the oscillation does, its time-averaged density equals the density at the equilibrium: the glide is not merely the bottom of the well but the *mean* of every oscillation about it. One consequence is used below: since $\langle m_\theta \rangle$ does not depend on W , the averaged motion of the slow state does not either, so the amplitude is an output of the reduced description and not a driver of it.

3.2 What ε does, and what it does not

Lemma 3.4 (ε is a clock and nothing else). *At frozen (c, θ) and $\kappa = 0$, the orbit of Eqs. (2.1) to (2.2) through amplitude W is, as a curve in the (z, v) plane, independent of ε ; and every time measured along it is $\sqrt{\varepsilon}$ times a quantity depending on (W, c, θ) alone. In particular the period and the time spent below the section $z = 0$ are*

$$P = \sqrt{\varepsilon} P_0(W, c, \theta), \quad P_{\text{dip}} = \sqrt{\varepsilon} D_0(W, c, \theta), \quad (3.5)$$

so that the dip fraction D_0/P_0 is free of ε .

Proof. The orbit is the level set $\frac{1}{2}v^2 = W - \Phi_{c,\theta}(z)$, in which ε does not appear. Along it Eq. (2.1) gives $d\tau = \sqrt{\varepsilon} dz/v$, so every elapsed time is $\sqrt{\varepsilon}$ times an integral over z of a function of (W, c, θ) . Taking the integral over the whole orbit and over its part with $z \leq 0$ gives Eq. (3.5). $\square$

This is why the three regimes can be separated by conditions that do not mix: the *shape* of an oscillation is set by its amplitude, and ε only decides how fast it is traversed. Fix two thresholds $\vartheta_1, \vartheta_2 \in (0, 1)$ and call a orbit

- a *bubble* if its dip fraction satisfies $D_0/P_0 \leq \vartheta_1$: the dip is short against the arc, so the pass is an event;
- *fast* if its period satisfies $\sqrt{\varepsilon} P_0 \leq \vartheta_2$: the oscillation is quick against the slow drift, so it may be averaged;
- *intermediate* otherwise.

Lemma 3.5 (When the intermediate band is empty). *D_0/P_0 is decreasing in W and P_0 is increasing in W , so the bubbles are the orbits with $W \geq W_1(\vartheta_1)$ and the fast orbits are those with $W \leq W_2(\vartheta_2, \varepsilon)$, where W_2 is increasing in ϑ_2 and in $1/\varepsilon$. The band is empty, and every orbit is a bubble or fast, as soon as $W_2 \geq W_1$, which holds for*

$$\varepsilon \leq \varepsilon_1 := \frac{c_{\max} \vartheta_1 \vartheta_2^2}{8}. \quad (3.6)$$

Remark 3.6 (The band is a finite- ε phenomenon). Equation (3.6) is the precise sense in which the difficulty of the application is quantitative rather than structural. At a fixed physical ε the band is empty only for thresholds so large that neither approximation is accurate, and it empties as $\varepsilon \rightarrow 0$ for any fixed pair. Theorem 4.2 is therefore stated under Eq. (3.6); what to do at a fixed ε is Remark 4.3, and it costs no new error term.

3.3 The bubble

Definition 3.7 (Pass map). For a orbit that is a bubble, let $\zeta^- = (v^-, s^-)$ be its data where it last crosses $z = 0$ downward and $\zeta^+ = (v^+, s^+)$ where it next crosses upward, computed from Eqs. (2.1) to (2.3) with $\varepsilon = 0$ in Eq. (2.3) — that is, with the slow state frozen except for the density term. The *pass map* is

$$J(v^-, s^-) = \{ (v^+, s^+) : \text{measurable } c(\cdot), \theta(\cdot) \text{ on the dip} \}.$$

Lemma 3.8 (The bubble is bounded, in closed form). *Let $v^- \leq -v_0 < 0$. Along every orbit of Definition 3.7: $\frac{1}{2}(v^+)^2$ and $\frac{1}{2}(v^-)^2$ both equal $\int_{z_-}^0 \Phi'_{c,\theta}$ up to the drift of (c, θ) across the dip, so*

$$\sqrt{\frac{c_{\min} m_-}{c_{\max} m_+}} |v^-| \leq v^+ \leq \sqrt{\frac{c_{\max} m_+}{c_{\min} m_-}} |v^-|;$$

the dip reaches no lower than the z_- with $\Phi_{c_{\min}, \theta}(z_-) = W$; its impulse is $P_{\text{dip}} \langle m_\theta \rangle_{\text{dip}}$ and is at most M ; and $|s^+ - s^-| \leq \Gamma_{\max} M$.

Proof. On the dip the orbit is the level set of Eq. (2.5), so $\frac{1}{2}v^2 = W - \Phi_{c,\theta}(z)$ at every point, and evaluating at the two crossings of $z = 0$ gives $\frac{1}{2}(v^\pm)^2 = W - \Phi_{c,\theta}(0)$ with the same W up to Eq. (3.1) integrated over the dip. The displayed inequalities are that identity with (c, θ) ranged over their bounds, using $\Phi_{c,\theta}(0) = 0$ and Assumption 2.1. The depth is where the level set meets $v = 0$, which by Lemma 2.7 is z_- , and is monotone in c . The impulse is $\int_{\text{dip}} m_\theta d\tau$, bounded by Assumption 2.3, and $s^+ - s^- = \int_{\text{dip}} m_\theta \Gamma$. $\square$

Lemma 3.9 (The pass map is Lipschitz). *With $\kappa_1 = \kappa_\Gamma$ and $I_{\max} = M$, there is $\text{Lip}_J \leq e^{\kappa_1 I_{\max}} (1 + (\Gamma_{\max} + m_+) I_{\max}/v_0)$ such that J is Lipschitz with that constant in the Hausdorff distance, for entry data with $|v^-| \geq v_0$.*

Proof. Parametrise the dip by the delivered impulse $\mu = \int m_\theta d\tau \in [0, I]$. In that variable $ds/d\mu = \Gamma(s, c, \theta)$ and $dv/d\mu = c$, neither involving z , both Lipschitz in s with constant κ_1 over a horizon $I \leq I_{\max}$; Gronwall gives $e^{\kappa_1 I_{\max}}$ on the data. The exit is where the level set returns to $z = 0$, at which $|dz/d\mu| = |v|/m_\theta \geq v_0/m_+$, so the exit impulse is Lipschitz in the data with constant m_+/v_0 , and carrying either orbit between the two exits moves its data by at most $(\Gamma_{\max} + c_{\max})$ times that. Taking suprema over (c, θ) gives the Hausdorff statement. $\square$

4 The limit system, and the theorem

Definition 4.1 (The limit system). Fix $v_0 > 0$ and thresholds ϑ_1, ϑ_2 . The limit system $\mathcal{H}$ has state (z, v, s) on the arcs and (s) on the glide, with three transitions.

Arc (flow). $\sqrt{\varepsilon} \dot{z} = v$, $\sqrt{\varepsilon} \dot{v} = -1$, $\dot{s} = F(s, c, \theta)$, while $z \geq 0$ and the air is not felt.

Bubble (jump). At $z = 0$ with $v \leq -v_0$ and dip fraction at most ϑ_1 , the state is replaced by $(0, v^+, s^+)$ with $(v^+, s^+) \in J(v^-, s^-)$, consuming no time.

Glide (averaged flow). Where the orbit is fast, $\dot{s} = F(s, c, \theta) + \frac{1}{c} \Gamma(s, c, \theta)$, by Eq. (3.3), and z is not carried.

$\mathcal{R}_\mathcal{H}(t)$ is the set of slow states reached at time t from $\mathcal{X}_0$ over all admissible c, θ and all selections from J , and $\mathcal{R}_\mathcal{H}(E) = \bigcup_{t \in E} \mathcal{R}_\mathcal{H}(t)$.

Note what the glide mode does *not* carry: by Eq. (3.3) the averaged slow field is independent of the amplitude, so no action coordinate is needed. The amplitude is an output of the reduced description.

We measure the residual in $|(z, v, s)| = |z| + |v| + |s|$ and write $\mathbb{B}(\varrho)$ for the corresponding ball, $\oplus$ for the Minkowski sum. Set

$$C_f = F_{\max} + m_+ \Gamma_{\max}, \quad \kappa_0 = \kappa_F + \frac{\kappa_\Gamma}{c_{\min}}, \quad \kappa_1 = \kappa_\Gamma, \quad G = e^{\kappa_0 T + (\kappa_1 + c_{\max}) M}, \quad (4.1)$$

and let C_A be the averaging constant of Lemma 4.4.

Theorem 4.2 (Outer inclusion with an explicit residual). *Assume Assumptions 2.1 to 2.5 and let $\vartheta_1, \vartheta_2 \in (0, 1)$ satisfy the band condition Eq. (3.6). Then for every $\varepsilon \in (0, \varepsilon_1]$,*

$$\mathcal{R}_\varepsilon(T) \subseteq \underbrace{\mathcal{R}_\mathcal{H}([T - \varrho, T + \varrho])}_{\text{skeleton, bubbles and glide}} \oplus \underbrace{\mathbb{B}(\varrho)}_{\text{residual}}, \quad \varrho = G(C_f c_{\max} M \vartheta_1 + C_A \vartheta_2). \quad (4.2)$$

Choosing the thresholds to minimise ϱ subject to Eq. (3.6) gives

$$\varrho = 3G \left(\frac{2C_f c_{\max} M C_A^2}{c_{\max}} \right)^{1/3} \varepsilon^{1/3}. \quad (4.3)$$

All constants are those of Assumptions 2.1 to 2.5 through Eq. (4.1); none depends on the control, the selection or the initial point.

Three remarks on the statement. The inclusion is over a short interval of times, of half-width the residual, because a dip of positive duration is compared with an instantaneous jump. The rate $\varepsilon^{1/3}$ is what this construction gives and is not claimed sharp; it is poorer than the rate for concentration alone because the same theorem also covers the averaged regime, and the two are traded off through Eq. (3.6). And the residual is *linear in each threshold* with no interior optimum of its own — the optimisation in Eq. (4.3) comes entirely from the band condition.

Remark 4.3 (At a fixed ε : the band, and what it costs). If Eq. (3.6) fails — which by Remark 3.6 is the case at a fixed physical ε for any accurate pair of thresholds — the intermediate orbits are neither jumped nor averaged but *propagated exactly*: they carry the fast pair (z, v) and the unapproximated field Eqs. (2.1) to (2.3). This adds no term to ϱ , because nothing is approximated. What it costs is dimension and amplification: the reduced problem carries $2 + d$ coordinates instead of d during those windows, and G acquires the factor $e^{\kappa_0 T_E}$ for their total duration T_E , which is at most $c_{\max} M$ by Eq. (3.4) *whatever the thresholds*. So the thresholds trade sharpness against dimension, not sharpness against sharpness.

Lemma 4.4 (Averaging over one orbit, with its constant). *Let s solve Eq. (2.3) along a fast orbit of period P and let $\bar{s}$ solve $\dot{\bar{s}} = F(\bar{s}, c, \theta) + \frac{1}{c}\Gamma(\bar{s}, c, \theta)$ with $\bar{s}(0) = s(0)$. Then on $[0, T]$*

$$|s - \bar{s}| \leq C_A P, \quad C_A = \frac{1}{2} e^{\kappa_0 T} \text{osc} (\Gamma_{\max} + \kappa_\Gamma C_f T),$$

where $\text{osc} = \max_{\text{orbit}} |m_\theta - 1/c|$.

Proof. Put $e = s - \bar{s}$. Subtracting, $\dot{e} = [F(s) - F(\bar{s})] + m_\theta[\Gamma(s) - \Gamma(\bar{s})] + (m_\theta - \frac{1}{c})\Gamma(\bar{s})$, so $|\dot{e}| \leq \kappa'_0 |e| + |(m_\theta - \frac{1}{c})\Gamma(\bar{s})|$ with $\kappa'_0 = \kappa_F + m_+ \kappa_\Gamma$. Let $R(\tau) = \int_0^\tau (m_\theta - \frac{1}{c})$. By Eq. (3.3) the integrand has zero mean over each period, so R is periodic with period P and $|R| \leq \frac{1}{2} P \text{osc}$. Integrating the last term by parts,

$$\left| \int_0^\tau (m_\theta - \frac{1}{c}) \Gamma(\bar{s}) \right| \leq |R| \Gamma_{\max} + |R| \int_0^\tau \left| \frac{d}{d\tau} \Gamma(\bar{s}) \right| \leq \frac{1}{2} P \text{osc} (\Gamma_{\max} + \kappa_\Gamma C_f T),$$

using $|\dot{\bar{s}}| \leq C_f$. Gronwall's inequality in integrated form then gives the claim. $\square$

5 Proof of Theorem 4.2

Fix $\varepsilon \in (0, \varepsilon_1]$, an admissible trajectory $x_\varepsilon = (z, v, s)$ on $[0, T]$ with control $c(\cdot)$ and selection $\theta(\cdot)$, and $x_\varepsilon(0) \in \mathcal{X}_0$.

Step 1: the two budgets

By Assumption 2.3, $\frac{d}{d\tau} \Theta_1(s) = \nabla \Theta_1 \cdot F + m_\theta \nabla \Theta_1 \cdot \Gamma \leq -c_1 m_\theta$, so

$$\int_0^T m_\theta(z) d\tau \leq M, \quad \text{and likewise} \quad \int_0^T \sqrt{m_\theta(z)} d\tau \leq Q. \quad (5.1)$$

Both hold for every control, every selection and every ε .

Step 2: the motion is a sequence of orbits, and how many

By Lemma 2.7 the frozen motion is a single closed orbit through each amplitude, and by Lemma 3.1 the amplitude changes only through the damping and the control, with the *a priori* bound Eq. (3.2). So $[0, T]$ decomposes into arcs, on which m_θ is not felt, and orbits. By Lemma 3.3 each orbit's impulse is its period divided by c , so by Eq. (5.1) the total time in orbits is at most $c_{\max} M$, and the total impulse is at most M .

Counting the bubbles: a orbit that is a bubble has dip fraction at most ϑ_1 , so its period is at least $P_{\text{dip}}/\vartheta_1$ and its impulse at least $P_{\text{dip}}/(c_{\max} \vartheta_1)$. Summing over bubbles and using Eq. (5.1),

$$\sum_{\text{bubbles}} P_{\text{dip}} \leq c_{\max} \vartheta_1 M. \quad (5.2)$$

This is the one estimate that makes the construction work, and it is worth reading twice: the number of bubbles grows like $\varepsilon^{-1/2}$, since each has impulse of that order, while each dip shortens like $\varepsilon^{1/2}$, and Eq. (5.2) says the product is bounded with no ε at all.

Step 3: every orbit is a bubble or fast

By Lemma 3.4 the dip fraction and the rescaled period depend on (W, c, θ) alone, and by Lemma 3.5 the band condition Eq. (3.6), which holds because $\varepsilon \leq \varepsilon_1$, makes the intermediate set empty. So each orbit is a bubble or fast, and the two treatments of Definition 4.1 exhaust the motion.

Step 4: a bubble is a jump, up to the drift over its dip

Let a bubble occupy $[\alpha, \beta]$ with $z(\alpha) = z(\beta) = 0$. By Definition 3.7 the exit data of the ε -free inner problem from $(v(\alpha), s(\alpha))$ differ from $(v(\beta), s(\beta))$ only through the terms dropped in Eq. (2.3), whose integral over the dip is at most $C_f P_{\text{dip}}$. Hence

$$(v(\beta), s(\beta)) \in J(v(\alpha), s(\alpha)) \oplus \mathbb{B}(C_f P_{\text{dip}}), \quad (5.3)$$

and $z(\alpha) = z(\beta) = 0$ exactly, so the altitude contributes nothing.

Step 5: a fast orbit is the averaged glide, up to $C_A P$

This is Lemma 4.4, with $P \leq \vartheta_2$ by the definition of a fast orbit, so the error is at most $C_A \vartheta_2$. Note which averaging this is. The model has one fast phase, so the average is over a single closed orbit, there is no resonance condition to satisfy, and the constant is the one computed in Lemma 4.4. Averaging over *several* phases, whose frequencies may cross, is a different problem and is not attempted anywhere in this document; it is the subject of objective 3, where we propose not to attempt it at all.

Step 6: assembling, and why the amplification is bounded

Construct $x_{\mathcal{H}}$ by following $\mathcal{H}$ with the same (c, θ) , jumping at each bubble with the point of J nearest to x_{ε} 's exit and flowing with the averaged field on each fast orbit. On the arcs and the glide both obey fields that are Lipschitz with constant κ_0 , so Gronwall gives the factor $e^{\kappa_0 T}$ over the horizon. At the bubbles the comparison passes through J , and Lemma 3.9 gives a factor $e^{(\kappa_1 + c_{\max})I}$ for a bubble of impulse I ; the product over all bubbles is therefore at most

$$\prod_{\text{bubbles}} e^{(\kappa_1 + c_{\max})I_k} = e^{(\kappa_1 + c_{\max}) \sum_k I_k} \leq e^{(\kappa_1 + c_{\max})M}$$

by Eq. (5.1). This is the second place the budget does work that no compactness argument could: the amplification compounds over $O(\varepsilon^{-1/2})$ bubbles and is nonetheless bounded, because each factor is exponential in an impulse and the impulses are budget-limited. Together the two factors are the G of Eq. (4.1).

Collecting Eqs. (5.2) and (5.3) for the bubbles and Step 5 for the fast orbits,

$$|x_{\varepsilon}(T) - x_{\mathcal{H}}(\hat{T})| \leq G \left(C_f \sum_{\text{bubbles}} P_{\text{dip}} + C_A \vartheta_2 \right) \leq G(C_f c_{\max} M \vartheta_1 + C_A \vartheta_2),$$

which is the ϱ of Eq. (4.2). The arrival time $\hat{T}$ differs from T because x_{ε} spends the dips where $x_{\mathcal{H}}$ spends no time, and by Eq. (5.2) that discrepancy is at most $c_{\max} \vartheta_1 M \leq \varrho / (G C_f) \cdot C_f \leq \varrho$, so $\hat{T} \in [T - \varrho, T + \varrho]$ and Eq. (4.2) follows.

Finally Eq. (4.3). Minimising $C_f c_{\max} M \vartheta_1 + C_A \vartheta_2$ subject to $\vartheta_1 \vartheta_2^2 = 8\varepsilon / c_{\max}$ — the boundary of Eq. (3.6), where the band is just empty and the thresholds are as small as it permits — gives $\vartheta_1 \propto \varepsilon^{1/3}$, $\vartheta_2 \propto \varepsilon^{1/3}$ and the stated value. $\square$

Remark 5.1 (Where each hypothesis was used). Assumption 2.1 was used once, in Lemma 2.7, and everything structural descends from it: one well, one dip per period, and hence countable passes. Assumption 2.3 was used twice, in Step 1, and everything quantitative descends from it: the pass count Eq. (5.2), the total time in orbits Eq. (3.4), and the bounded amplification of Step 6. Assumption 2.4 was used once, for the *a priori* amplitude bound Eq. (3.2). Assumptions 2.2 and 2.5 supply the constants. No compactness argument and no measure appears anywhere above.

6 The weak formulation

[Theorem 4.2](#) is a statement about trajectories, and its proof uses no measures. The weak formulation is nevertheless where the question belongs, for two reasons: it is what makes the solution set of a nonconvex family a tractable object at all, by rendering the dynamics a linear constraint; and it is the setting in which the loss of compactness that motivates the whole decomposition is visible. We give the definitions, so that what is open can be stated precisely.

For a trajectory of [Eqs. \(2.1\) to \(2.3\)](#) with control $c(\cdot)$ and selection $\theta(\cdot)$, the *occupation measure* is the element $\mu_\varepsilon \in \mathcal{M}_+([0, T] \times \mathcal{X} \times \mathcal{U} \times \Theta)$ defined by

$$\int \varphi d\mu_\varepsilon = \int_0^T \varphi(t, x_\varepsilon(t), c(t), \theta(t)) dt \quad \text{for } \varphi \in C([0, T] \times \mathcal{X} \times \mathcal{U} \times \Theta),$$

together with the initial and terminal measures μ_0, μ_T carried by $x_\varepsilon(0)$ and $x_\varepsilon(T)$. The topology is the weak-* topology, that is, the topology of the dual of that space of continuous functions; the base is a compact metric space by [Assumption 2.2](#), so the topology is metrizable and bounded sets are relatively compact. In these terms the dynamics are a *linear* constraint: for every $\varphi \in C^1$,

$$\int \varphi(T, \cdot) d\mu_T - \int \varphi(0, \cdot) d\mu_0 = \int \left(\frac{\partial \varphi}{\partial t} + \nabla \varphi \cdot (f_0 + \rho_\varepsilon b) \right) d\mu_\varepsilon, \quad (6.1)$$

which is the Liouville equation of the family, and the reachable set at time T is the union of the supports of the terminal measures that satisfy it. This is the occupation-measure program of Lasserre, Henrion and collaborators [\[5, 6, 7\]](#), and an outer bound comes from its dual: a pair (Φ, Ψ) with Φ non-decreasing along the admissible motions and at least one on $\mathcal{X}_0$, and Ψ majorising Φ at time T , encloses the reachable set in $\{\Psi \geq 1\}$.

What [Theorem 4.2](#) does is to hand this machinery a different problem. After the theorem, the object to be enclosed is $\mathcal{R}_\mathcal{H}$, which is a reachability problem in $2 + d$ coordinates with a flow, one guard $y = 0$, a set-valued reset J , and at most N_* jumps. That is exactly the hybrid form for which the occupation-measure relaxation is known [\[8, 9\]](#): a measure on each guard carries the mass from one mode to the next, and the certificate Φ must be non-decreasing along the arc flow and across the reset. The singular limit has been discharged by [Section 5](#), which is elementary, and what is left on $\mathcal{H}$ is a problem in a few coordinates. Whether the limit is better taken in this topology, with the decomposition extracted from the measures, or at the level of trajectories as in [Section 5](#), is the open question of [Section 1.1](#), and the two routes are not known to be equivalent.

Two things are not done. [Theorem 4.2](#) compares trajectories and not measures, and the companion statement at the level of measures is [objective 4](#). And nothing above makes $\mathcal{R}_\mathcal{H}$ tight: an outer bound on it is an outer bound on $\mathcal{R}_\varepsilon(T)$ up to the residual, by [Eq. \(4.2\)](#), but how much is lost in passing from the one to the other is a separate question from how much is lost in the limit.

7 What hypersonic entry adds

The application is hypersonic entry: the reachable set of an entry vehicle, for any vehicle and any corridor, with the vehicle and its corridor entering only as data. Entry is not an arbitrary illustration, because it supplies the singular perturbation, the limit system and the concentrating events in one system. Nor is it already covered: the exact method solves a Hamilton–Jacobi–Isaacs equation on a grid, capped in practice near four or five states [\[10, 11\]](#), while a rigid-body entry vehicle carries thirteen and more once ablation and the aerothermoelastic modes are coupled [\[12\]](#); and set-propagation methods return rigorous enclosures at ten to twenty states [\[13, 14, 15\]](#), but of one instance — one vehicle, one initial box, one horizon — with no explicit dependence on the corridor or on the width of the aerodynamic uncertainty, which is what a statement uniform over a family requires.

7.1 The correspondence

Measure altitude in scale heights above the equilibrium glide, so that the atmospheric density is $m_\theta(z)$ with m_θ decreasing, and let

$$\varepsilon = H_s/R_e$$

be the thin-atmosphere ratio, whose square root is the ratio of the oscillation period about the glide to the slow time $\sqrt{R_e/g}$. Then Eqs. (2.1) to (2.2) are the vertical motion of the vehicle: v is the rate of climb in those units, c is the vertical lift fraction $(L/D)\cos\sigma$ normalised so that $cm_\theta = 1$ is the glide balance — lift against weight less centrifugal relief — and κ is the phugoid damping, which does not scale with ε . The slow state s collects the energy, the range and the heading, F carries the gravitational terms and Γ the drag. The selection θ chooses the density shape and the aerodynamic coefficients from an enclosure, so that Eqs. (2.1) to (2.3) are a differential inclusion rather than a single field [16, 17], and Remark 2.6 is the statement that the bound holds over the whole enclosure. Here the vehicle enters only through dimensionless ratios — c , κ , the budget constants — each quantified over a range for a class, never evaluated at a design.

The identification of the three regimes is the content of the model rather than an analogy. The minimum of the well is the equilibrium glide; the small oscillation about it is the phugoid, of period $\sqrt{\varepsilon}$ on the slow time scale and damping ratio of the same order, so that it persists for the whole entry; and a large oscillation is a skip, rebounding from the exponential wall in a short time and coasting on a long ballistic arc. That the dip is short and the arc long is Lemma 3.4 together with the asymmetry of $\Phi_{c,\theta}$, and nothing more.

Assumption 2.3 is the hypothesis that transfers exactly. The specific mechanical energy in the rotating frame, $\mathcal{E} = \frac{1}{2}V^2 + \Phi_g - \frac{1}{2}\omega_E^2 r^2 \cos^2\phi$, satisfies $\dot{\mathcal{E}} = -P$ with $P = DV/m \geq 0$ the drag power, because neither the lift nor the Coriolis force does work. Taking $\Theta_d = \mathcal{E}$ gives $\nabla\Theta_d \cdot F = 0$ exactly, and $\nabla\Theta_d \cdot \Gamma \leq -c_0$ with c_0 fixed by lower bounds on the drag coefficient and the speed, since the drag is strictly proportional to the density. The budget, the pass count and the bound on the time spent in the layer therefore hold for entry as they stand. Assumption 2.1 holds on a corridor bounded away from circular speed, where $g - V^2/r$ is bounded below, and with the bank bounded away from ninety degrees; Assumption 2.2 holds away from the geographic poles, where the spherical chart is singular.

7.2 Why the band is not empty

The hypothesis entry does not satisfy is the band condition Eq. (3.6), and the reason is quantitative. A pass entered at a flight-path angle γ_e crosses a scale height in a time of order ε/γ_e on the slow time scale and levels out under a load of order γ_e^2/ε relative to gravity, so the entry angle enters through the ratio $\gamma_e/\sqrt{\varepsilon}$, as in the classical Chapman–Loh variables [18, 19]. For the Earth $\sqrt{\varepsilon} \approx 0.034$, about two degrees, and entry angles are a few degrees: a pass is therefore neither short compared with the arc that follows it nor distinct in scale from the oscillation about the glide. In the language of Lemma 3.5, ε exceeds ε_1 for every pair of thresholds small enough to be accurate, so the intermediate band is not empty and Remark 4.3 rather than Theorem 4.2 is what applies. That costs no term in the residual; it costs dimension, over a total duration bounded by Eq. (3.4).

8 Plan of work

Theorem 4.2 is the first of the obligations, and it is discharged. Four remain. We state each with the physical obstruction it has to overcome.

Objective 1. Theorem A: the profile decomposition, with a constant

We must prove the decomposition itself: that every sequence of trajectories with $\varepsilon_n \rightarrow 0$ admits, along a subsequence, a splitting into a skeleton, finitely many profiles obtained by rescaling the passes to their own widths, and a remainder small in a stated norm. The classical route iterates the dichotomy of the concentration-compactness principle [1, 20] and, in the form that resolves each profile at its own scale, extracts a pairwise orthogonal family [2, 21]; orthogonality is produced by the construction rather than assumed, and profiles at distinct times are orthogonal here with nothing to prove, since the scale of a profile is the width of a pass and its centre the time at which the pass occurs.

What we expect to add is the constant. In that theory the number of profiles above a given size is finite because the conserved quantity is finite, and the argument is by contradiction; here [Assumption 2.3](#) supplies a Lyapunov functional whose total variation is bounded uniformly in ε , so the count is bounded *in advance* and explicitly, as [Eq. \(5.2\)](#) already shows for the dips. The objective is met by the decomposition stated with that count, with the remainder measured in the largest share of dissipation held by any window of fixed rescaled width — the norm in which a remainder of this kind is small, and in which it is not small in the conserved quantity [2, p. 136].

Objective 2. The statement at a fixed ε , and a smaller selection class

[Theorem 4.2](#) holds under the band condition [Eq. \(3.6\)](#), which the application fails. We must prove the statement [Remark 4.3](#) describes: the intermediate orbits carried by the unapproximated field, the reduced problem acquiring the fast pair over a total duration bounded by [Eq. \(3.4\)](#), and the residual gaining no new term. The enclosure of an intermediate orbit is a propagation over a bounded time rather than an asymptotic estimate, so what has to be controlled is its growth, not its order; the obligation is a bound on that growth which is explicit in the interval widths.

The same objective carries the one refinement that would most sharpen the result. In [Assumption 2.2](#) the selection $\theta(\cdot)$ is an arbitrary measurable function: its range is bounded and its variation is not. The atmosphere is smooth in space, mass is lost monotonically, and coefficient errors are correlated, so the admissible class is far smaller than an arbitrary selection, and a bound proved over the smaller class is tighter at no cost to validity. We must state that class — bounded variation in altitude, monotonicity where it holds — and carry it through [Lemmas 3.8](#) and [3.9](#). The objective is met by the fixed- ε theorem with the enclosure growth explicit, over the restricted class.

Objective 3. The remainder: motions that do not concentrate

A profile decomposition separates what concentrates from what does not, and the second part is the remainder. The entry problem has motions of that kind which Eqs. (2.1) to (2.3) does not carry: the attitude and the elastic modes oscillate for as long as there is air, and nothing about them concentrates, so in the decomposition they are not profiles at all. The wave-equation theory is careful to say that its remainder is not small in the conserved quantity, since it contains every motion that concentrating profiles cannot represent [2, p. 136], and the question is therefore in what sense the remainder *is* small here.

The classical answer is averaging [22], and for *several* phases it does not reach a bound. (Over a single orbit it does, and Lemma 4.4 uses it; the difficulty is entirely in the passage to more than one phase.) An averaging theorem gives an error to an order; with several phases it holds only away from resonance, while the attitude frequency grows with the square root of the dynamic pressure and so rises from zero at the start of every pass; and the results that carry a solution through a resonance hold outside a set of initial conditions of small measure, which an outer statement about a whole solution set cannot discard. We propose a Lyapunov functional in place of an averaging theorem — to bound what these motions *do* rather than to resolve what they are. They act on the translational motion only through the aerodynamic force, so it is the force that must be enclosed, and the enclosure may then be carried by the selection θ at no cost in dimension. Two facts are to be established. Where there is no air there is no torque, so the angular rates are bounded by conservation and the attitude has no aerodynamic effect in any case. Where there is air, the attitude damping rate is proportional to the product of density and speed, while the energy identity fixes the integral of density against the cube of speed, so the budget that bounds the dissipation also bounds the integrated damping from below — the same air does both. The objective is met by a bound on the aerodynamic force error, explicit in the dimensionless groups of the class and uniform over the corridor, with the elastic modes included or enclosed as a widened selection set.

Objective 4. The measure-level companion and the certificate

We must show that the occupation measures of Section 6 converge to measures satisfying the Liouville equation of the limit system, with the mass crossing each pass carried by a measure on the graph of J , uniformly over the admissible family, and exhibit the dual pair (Φ, Ψ) non-decreasing along the arc flow and across the reset. This is what makes Theorem 4.2 a statement about the limit of the family rather than a comparison of two trajectories, and it is where weak-* compactness and concentration-compactness in the form that resolves each profile at its own scale [1, 20] are expected to be needed — for the structure, with the constants coming from the functionals of Assumption 2.3 as in objective 1.

Objective 5. Validated evaluation of the certificate

The guarantee is to be verified in validated arithmetic rather than produced by a numerical solver, and this is the obligation set aside in Section 1.1. We must show that the closed-form constants, the enclosures of the passes they do not cover, and the reachable set of the limit system can be evaluated over the whole corridor with every rounding error accounted for. Two losses have to be managed and neither is removed by refining a step: repeated occurrences of a variable are treated as independent, and the image of a box under a flow is not a box, so re-enclosing it compounds [3, Chs. 4–5, 10]. What controls the second is the representation of the set [4, 15, 23]. The objective is met when every inequality the certificate asserts is discharged by a computation returning either a proof or an explicit failure to decide at the working precision.

8.1 Sequencing

Three phases rather than four parallel tracks. The first extends Theorem 4.2 to the planar entry problem, where the correspondence above is exact and Assumptions 2.1 and 2.3 are verified rather than assumed; the small question of whether the logarithm in $\varrho(\varepsilon)$ can be removed belongs to it. The second is objective 2, in which the thresholds of Lemma 3.5 and the selection class are the two dials on the same statement. The third

is [objective 3](#), and it should be reached early enough that a negative result is still recoverable; [objective 4](#) runs alongside it, needing only that the limit system be settled. If [objective 3](#) resists a proof where frequencies cross, the retreat is a corridor on which the frequency bands are ordered, with the crossings at the edges of each pass left to the enclosure of the bubble, which integrates the coupled field and needs no separation; that weakens the residual without invalidating the decomposition.

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